

CHEMICAL CHANGES – ELECTROLYSIS

KEY VOCABULARY

Electrolysis: using electricity to break down an ionic compound.

Electrolyte: the molten or dissolved ionic compound.

Cathode: the negative electrode — attracts positive ions.

Anode: the positive electrode — attracts negative ions.

COMMON MISCONCEPTIONS

~~Solid ionic compounds can be electrolysed.~~

✓ Ions must be free to move — molten or dissolved.

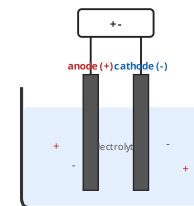
~~Positive ions go to the anode.~~

✓ Positive ions go to the negative cathode.

~~Electrons flow through the electrolyte.~~

✓ Ions carry charge in the electrolyte, not electrons.

ELECTROLYSIS CELL



1 WHY MOLTEN

Explain why an ionic compound must be molten or dissolved to be electrolysed. [2 marks]

2 ELECTRODE NAMES

State which electrode is the **cathode** and its charge. [1 mark]

3 PRODUCTS (MOLTEN)

Molten lead bromide is electrolysed. Name the product at each electrode. [2 marks]

4 HALF EQUATION

Write the **half equation** for lead ions at the cathode. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Positive ions move to the ____ and ____ ions move to the anode.

cathode

anode

positive

negative

6 OXIDATION/REDUCTION

Explain why reduction happens at the cathode. [2 marks]
